

A Tale of Five Decoders.

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October 31, 2025

Introduction

Today:

- ▶ Noisy Circuits.
- ▶ Definitions and Motivation.
- ▶ Pippenger Construction. (Classical, Fault Tolerance with constant overhead at depth).
- ▶ 'French-line' works, modern fault tolerance methods and gadgets. ('log n' overhead at depth).
- ▶ An almost $\mathbf{QNC}_1 = \text{noisy-}\mathbf{QNC}_1$.
- ▶ Next week, directions and hints that might show separation. ($\neq$).

TAKEAWAYS:

- ▶ More about codes.
- ▶ First view to fault tolerance.
- ▶ Nice open problems.

Map

How to Read

A

slide that explains how to parse the plots.

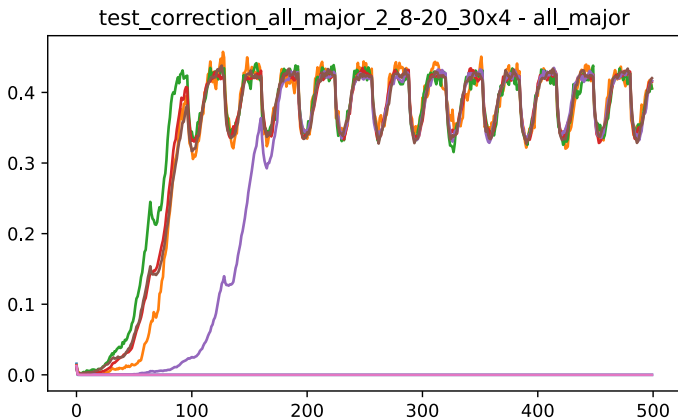
Ideal Result.

A

slide that explains what we would expect to get.

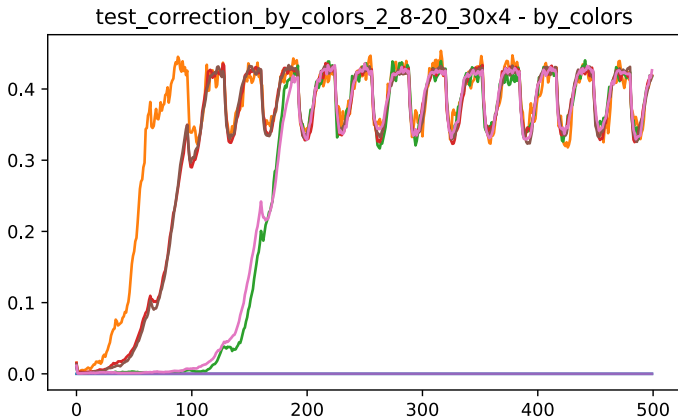
Ideal Result.

Majority



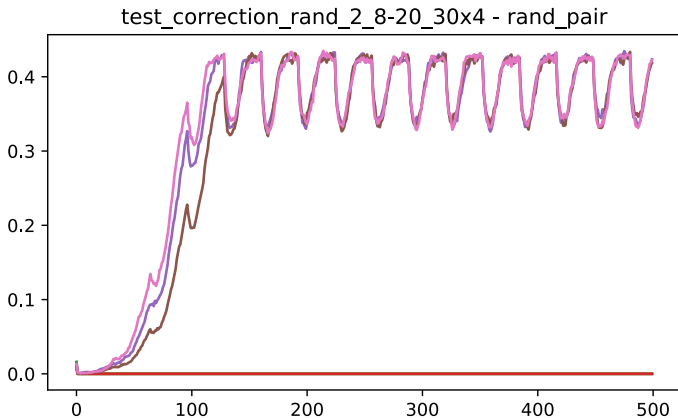
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Colors



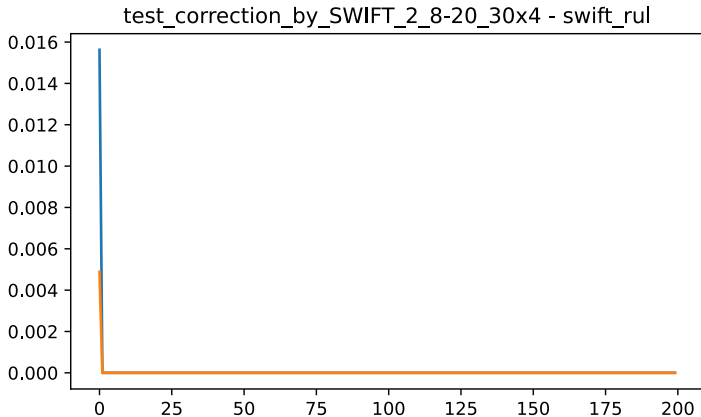
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Picking Random



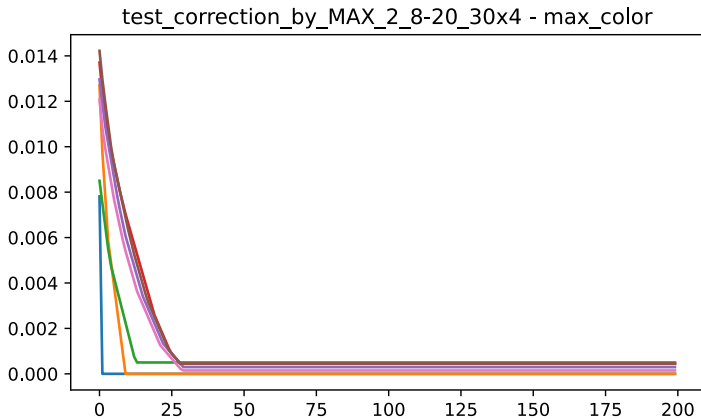
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

SWIFT



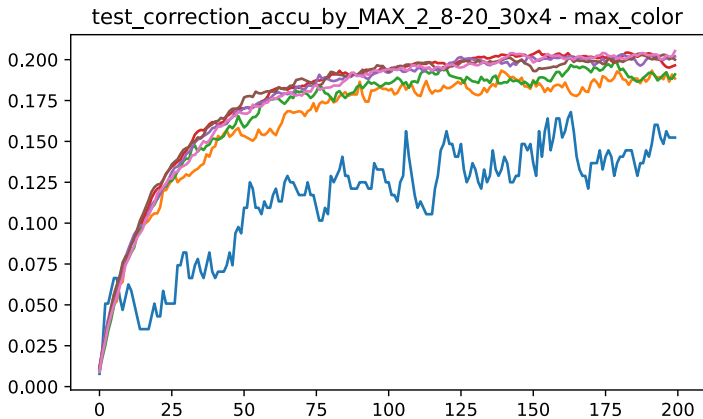
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True